

Midterm

CSED501

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Due: **November 11, 2025**

Name: _____

Instructions: Complete the questions below and upload a PDF containing your written answers to Gradescope.

- Keep your answers within the provided space.
 - Answers can be handwritten, or typed and pasted into this PDF.
 - Show your work; this makes giving partial credit easier!
 - Numeric answers can be given in fraction or decimal form; if decimal, round to 2 places
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Problem 1 8 pts:

True/False. Mark each statement as True or False and briefly justify your answer.

- (a) (2 pts) The variance of a model generally **decreases** as the number of features **increases**.
- (b) (2 pts) The variance of a model generally **decreases** as the weight of the regularization error **increases**.
- (c) (2 pts) Assume we train a model on a given dataset. If we were to remove 50% of samples from the dataset and re-train the model from scratch, the new model will be more likely to overfit to its training data than the old one.
- (d) (2 pts) A solution to a convex optimization problem is guaranteed to be a global minimum.

Answer:

Problem 2 24 pts:

Short Answer. For each problem, provide a short answer and explanation(1 paragraph). When providing ML methods, only provide ML methods that we have covered in class.

- (a) (8 pts) You have an unlabeled data set containing descriptions of two types of products, Product A and Product B. This description includes features such as color, weight, and other relevant information. You are interested in grouping the data to try to separate out Product A and Product B, but you don't have time to label the dataset by hand. What ML method(s) would you try, and why?
- (b) (8 pts) You work for a large clothing company, and your boss wants you to write a program that predicts whether a new item of clothing will sell well based on features such as price, material, and color. Your boss wants you to train a model fairly quickly, but is willing to purchase the compute needed to parallelize code. What ML method(s) would you try, and why?
- (c) (8 pts) Your coworker has been recording how long it takes them to finish coding tasks at work for the last year, and has noted features of each task such as the number of files that need to be written or edited, the number of people working on the task, the perceived difficulty of the task, and others. They ask you to write a program to predict how long a coding task will take them to finish. What ML method(s) would you try, and why?

Answer:

Problem 3 28 pts:

In Figure 1 We have two labeled datasets with two and three classes, respectively. Assume that within each dataset the classes are separable, albeit not necessarily linearly separable. Furthermore, assume that no two distances are equal and that we implement the algorithms below on a magical computer with no rounding error.

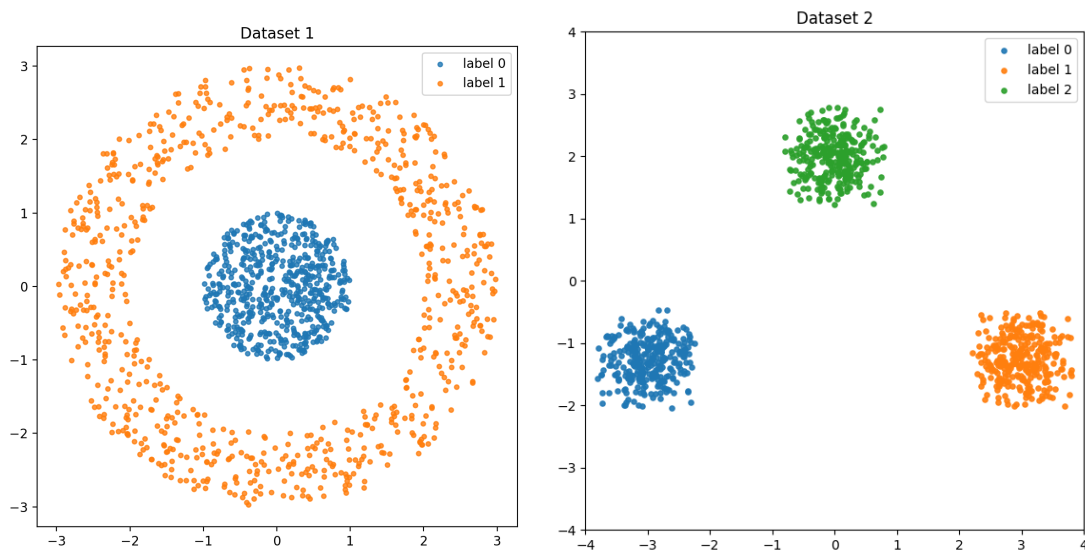


Figure 1: The two datasets. Colors correspond to labels.

- (2 pts) Which dataset is linearly separable and which one is not?
- (5 pts) For the dataset that is linearly separable, how many weight vectors would **softmax logistic regression** learn, and what are the dimensions of those weight vectors?
- (5 pts) For the dataset that is not linearly separable, describe a feature transformation that makes the data linearly separable in that feature space.
- (8 pts) For the assumptions in the intro, and $k = 2$ on the first dataset and $k = 3$ on the second dataset, is Agglomerative Clustering using single linkage **guaranteed** to correctly identify the clusters in each dataset (that is, are you guaranteed to be able to cut off Agglomerative Clustering using single linkage at a point that identifies the clusters shown)? Briefly explain why/why not.
- (8 pts) For the assumptions in the intro, and $k = 2$ on the first dataset and $k = 3$ on the second dataset, is K-means **guaranteed** to correctly identify the clusters in each dataset? Briefly explain why/why not.

Answer:

Problem 4 18 pts:

In this problem, we will show that the Perceptron algorithm can be interpreted as performing stochastic gradient descent with a batch size of 1 on a hinge-like loss. Recall that the Perceptron algorithm classifies examples (x_i, y_i) with $y_i \in \{-1, +1\}$ using a linear model $f(x; w) = w^\top x$ and the classification rule:

$$\hat{y}_i = \begin{cases} 1, & \text{if } f(x_i; w) \geq 0, \\ -1, & \text{if } f(x_i; w) < 0. \end{cases}$$

- (a) (5 pts) Write the Perceptron update rule, in vector form, for a point (x_i, y_i) and step size η . Write your answer in the following format:

$$w^{t+1} \leftarrow \begin{cases} \text{TODO}, & \text{if } y_i = \hat{y}_i, \\ \text{TODO}, & \text{if } y_i \neq \hat{y}_i. \end{cases}$$

- (b) (8 pts) Define a *hinge-like* loss function

$$\ell(w; x_i, y_i) = \max(0, -y_i \cdot f(x_i; w)).$$

We will compute the gradient $\nabla_w \ell(w; x_i, y_i)$. Note that the function $m(a) = \max(0, a)$ is not differentiable at $a = 0$. For simplicity, assume that

$$\frac{\delta m(a)}{\delta a} = \begin{cases} 0, & \text{if } a < 0, \\ 1, & \text{if } a \geq 0. \end{cases}$$

Let $a(w; x_i, y_i) = -y_i \cdot f(x_i; w)$. Using the chain rule,

$$\nabla_w \ell(w; x_i, y_i) = \frac{\delta m(a(w; x_i, y_i))}{\delta a} \cdot \nabla_w a(w; x_i, y_i).$$

Complete the derivation of $\nabla_w \ell(w; x_i, y_i)$ in the following case format (reminder: $\nabla_w a(w; x_i, y_i)$ is the gradient of $a(w; x_i, y_i)$ with respect to w).

$$\nabla_w \ell(w; x_i, y_i) = \begin{cases} \text{TODO}, & \text{if } a < 0, \\ \text{TODO}, & \text{if } a \geq 0. \end{cases}$$

- (c) (5 pts) Show that the Perceptron update from part (a) is equivalent to performing one stochastic gradient descent step for the point (x_i, y_i) with the same step size η :

$$w^{t+1} \leftarrow w^t - \eta \nabla_w \ell(w; x_i, y_i).$$

You should again express this as a two-case update as in part (a) to show that the updates are the same. Show your work.

- (d) (5 pts) **Extra credit:** Explain why the loss above is called “hinge-like” and describe how it differs from the actual SVM hinge loss. What advantage does the SVM hinge loss have over the loss $\ell(w; x_i, y_i)$ we defined?

Answer:

Problem 5 22 pts:

Suppose we have a dataset $\{X_1, \dots, X_n\}$ of size n where each sample is an i.i.d. (independent and identically distributed) random variable $X_i = \theta + \varepsilon_i$ for some fixed θ and random noise ε_i with $\mathbb{E}[\varepsilon_i] = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$. That is, each X_i starts with the same fixed θ value, and then adds random noise. We use three very simple estimators of θ to investigate the bias-variance trade off:

- **Estimate θ using only one sample:** $\hat{\theta}_1 = X_1$.
- **Sample mean:** $\hat{\theta}_2 = \frac{1}{n} \sum_{i=1}^n X_i$.
- **An estimator that is clearly wrong(or is it?):** $\hat{\theta}_3 = 0.5 \cdot X_1$.

We evaluate estimators by their mean squared error (MSE) on a fresh i.i.d. sample $X_{\text{new}} = \theta + \varepsilon_{\text{new}}$. Recall that for some estimator $\hat{\theta}$ of θ , we can split the MSE into three terms:

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(X_{\text{new}} - \hat{\theta})^2] = \underbrace{(\mathbb{E}[\hat{\theta}] - \theta)^2}_{\text{Bias}^2} + \underbrace{\text{Var}(\hat{\theta})}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible Error}}.$$

- (a) (4 pts) In your own words, what are the expectations in this formula taken over? That is, what is the source of randomness here?
- (b) (9 pts) For $\hat{\theta}_1$, compute the bias squared, variance, and MSE. Your final answers should only be in terms of σ and θ . Show your work. Note: the variance of a fixed value is 0, and $\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$, and if A and B are independent, $\text{Var}(A + B) = \text{Var}(A) + \text{Var}(B)$.
- **The Bias² term:** $\text{Bias}(\hat{\theta}_1)^2 = (\mathbb{E}[\hat{\theta}_1] - \theta)^2$.
 - **The Variance term:** $\text{Var}(\hat{\theta}_1)$.
 - **The MSE:** $\text{MSE}(\hat{\theta}_1)$.

- (c) (9 pts) We will compute the bias squared, variance, and MSE for $\hat{\theta}_2$ and $\hat{\theta}_3$ for you:

For $\hat{\theta}_2$:

- **The Bias² term:**
 $\text{Bias}(\hat{\theta}_2)^2 = (\mathbb{E}[\hat{\theta}_2] - \theta)^2 = (\mathbb{E}[X_1] - \theta)^2 = (\mathbb{E}[\frac{1}{n} \sum_{i=1}^n X_i] - \theta)^2 = (\theta + \frac{1}{n} \mathbb{E}[\sum_{i=1}^n \varepsilon_i] - \theta)^2 = (\theta - \theta)^2 = 0.$
- **The Variance term:** $\text{Var}(\hat{\theta}_2) = \text{Var}(\frac{1}{n} \sum_{i=1}^n X_i) = \text{Var}(\frac{1}{n} \sum_{i=1}^n \varepsilon_i) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(\varepsilon_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}.$
- **The MSE:** $\text{MSE}(\hat{\theta}_2) = \frac{\sigma^2}{n} + \sigma^2 = \frac{\sigma^2(n+1)}{n}.$

For $\hat{\theta}_3$:

- **The Bias² term:**
 $\text{Bias}(\hat{\theta}_3)^2 = (\mathbb{E}[\hat{\theta}_3] - \theta)^2 = (\mathbb{E}[0.5 \cdot X_1] - \theta)^2 = (0.5\mathbb{E}[X_1] - \theta)^2 = (0.5\theta - \theta)^2 = 0.25\theta^2.$
- **The Variance term:** $\text{Var}(\hat{\theta}_3) = 0.5^2 \text{Var}(X_1) = 0.25\sigma^2.$
- **The MSE:** $\text{MSE}(\hat{\theta}_3) = 0.25\theta^2 + 0.25\sigma^2 + \sigma^2 = 0.25\theta^2 + 1.25\sigma^2.$

QUESTION CONTINUED ON NEXT PAGE

Problem 5 TODO pts: CONTINUED FROM PREVIOUS PAGE

Using $\hat{\theta}_1$ as the baseline, briefly answer the following:

- How do $\hat{\theta}_2$ and $\hat{\theta}_3$ compare in terms of bias squared, variance, and overall MSE?
- Which estimator ($\hat{\theta}_2$ or $\hat{\theta}_3$) behaves most like a regularized regression model (e.g., Ridge or Lasso)?
- Is there anything we can do to reduce the irreducible error term?

Answer:

